

SAI VENKATA HARSHINI RAYAVARAPU

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SUMMARY

Software and Machine Learning Engineer with 14 months of production data engineering at Accenture and an MS in Computer Science (Cleveland State). Builds end-to-end: two-tower neural retrieval at sub-50ms p95, multi-agent LLM systems with automated SQL validation, and causal inference frameworks separating true treatment effect from correlation. Deep SQL/data-pipeline background on high-volume financial data.

TECHNICAL SKILLS

Languages: Python (PyTorch, Pandas, NumPy, Scikit-Learn), SQL, C++, COBOL / JCL

Machine Learning: Two-Tower Retrieval, Embeddings, FAISS Vector Search, Transformers, LLMs (Llama 3.3), Causal Inference, Propensity Score Matching, NLP (TF-IDF)

Data & Infrastructure: FastAPI (REST APIs), Docker, DuckDB, MLflow, Git, Power BI, Streamlit

Practices: MLOps, Multi-Agent Orchestration, Query Optimization, System Design, Counterfactual Analysis

PROJECTS

NeuralStream — Neural Retrieval & Ranking System

PyTorch | FAISS | FastAPI | Docker

- Built a two-stage retrieval-and-ranking pipeline: a two-tower network embeds users and items in a shared 64-D space.
- Built a FAISS flat inner-product vector-search index over a 1M-item catalog (embeddings precomputed offline), serving top-K retrieval at sub-50ms p95 latency.
- Built a containerized FastAPI service (Docker) to serve ranked results.

Starbit Engine — Autonomous Multi-Agent Data Intelligence

LangGraph | Llama 3.3 | DuckDB

- Architected a LangGraph multi-agent system that plans, writes, validates, and visualizes SQL against DuckDB without human intervention.
- Added a critic agent that catches hallucinated or invalid SQL and triggers bounded recursive retries, reducing failed queries on multi-table schemas.

Causal Analytics — Incremental Lift & ROI Framework

Scikit-Learn | Statsmodels | Pandas

- Built a causal inference pipeline measuring true incremental lift of product features, exposing a 7x over-estimation in the naive correlational estimate.
- Constructed counterfactual control groups via Propensity Score Matching (KNN), isolating treatment effect from self-selection bias in high-engagement user segments.
- Validated the estimator on a simulated campaign dataset with known ground truth before applying it to observational data.

PROFESSIONAL EXPERIENCE

Accenture Pvt Ltd — Bengaluru, India

Application Development Associate / Software Engineering Analyst | October 2022 - December 2023

- Built and maintained high-volume banking data pipelines in Mainframe SQL: ETL jobs and data-quality checks keeping downstream reporting consistent.
- Tuned SQL execution plans and indexing to reduce data retrieval latency across reporting queries.
- Replaced manual reconciliation steps with automated SQL pipelines, reducing recurring manual effort.
- Re-engineered legacy reporting logic into modern, auditable SQL with documented data-quality controls.

EDUCATION

Cleveland State University — Cleveland, OH

Master of Science in Computer Science | GPA: 3.4 / 4.0 | January 2024 - May 2026

Coursework: Artificial Intelligence, Machine Learning, Data Science, Network Security & Privacy

Vignan's Institute of Information Technology — Visakhapatnam, India

Bachelor of Technology in Computer Science & Engineering | GPA: 8.39 / 10.0 | May 2022

CERTIFICATIONS

OpenAI Academy — Agents and Workflows (2026) | Databricks Accreditation — AI Agent Fundamentals & Databricks Fundamentals (2026) | AWS — Fundamentals of ML & AI (2026)

Work Authorization: F-1 STEM OPT in process (36-month eligibility); no initial sponsorship required